

Trajectory plot

Forward Kinematics

1. **Fixed frame** — the crank center and the second fixed pivot sit a set distance A apart at angle θ_A , found from lengths a and l using basic right-triangle math.
2. **Two rigid triangular plates** — since a plate's shape never changes, its internal angles are solved once (law of cosines + sines) and kept as fixed constants (θ_{triE} , θ_{triH}).
3. **Crank drives two four-bars at once** — same crank, same frame, different lengths. Each is solved with the standard Freudenstein equation to get the rocker's angle — a direct formula, no guesswork.
4. **Locate points D and C** — take each rocker's angle (adding the fixed plate offset for D), walk out the link length, and you have both points.
5. **Measure the D–C gap** — call it S . Not a real link, just the current distance between two already-known points.
6. **Solve a new triangle using that gap** — sides S , F , G give you the location and orientation of a brand-new joint.
7. **Foot tip** — add the fixed plate offset θ_{triH} to get the foot arm's direction, then walk out distance l from C to land exactly on the foot.

Optimization

- **Free variables:** 13 link lengths (each scaled $0.70\times$ – $1.30\times$ off classical Jansen proportions) + crank phase (-180° to 180°) = 14 total.
 - **Optimizer:** Differential Evolution (pop 60, 500 gens, $F=0.65$, $CR=0.90$), seeded with the unscaled Jansen values, then polished with Nelder–Mead (fminsearch).
 - **Rescaling:** every trial's raw path is scaled so its height = 200 mm and shifted to match the target's position — so only *shape and phase* are searched; size/position are fixed afterward, not optimized.
 - **Objective:** weighted RMS error — full-path RMS + $4\times$ stance-only RMS (points near the ground) + smaller penalties for stride, peak-timing, and height mismatch. Lower = better fit.
- a,l-frame components (ground link O2–O4)
 - m-crank (O2 to crank pin)
 - j-coupler, loop 1 (crank pin to rocker 1)
 - b-rocker 1 (O4 to loop-1 output joint)

- k-coupler, loop 2 (crank pin to rocker 2)
- c-rocker 2 (O4 to loop-2 output joint, point C)
- d,e-internal sides of the rigid triangle on rocker 1, fixing dyad-attachment point D
- f,g-the two dyad links from D and C to the new foot-carrying joint
- h-internal side fixing the foot arm's angular offset on the dyad body
- i-foot arm — distance from C to the actual foot tip